

# **DBMS Basic to Advanced Notes**

## **Introduction to DBMS**

DBMS (Database Management System) is software used to store, manage, and retrieve data efficiently.

## **Advantages of DBMS**

Reduced redundancy, data consistency, security, backup, and sharing.

## **Database Users**

DBA, Application Programmers, End Users, Database Designers.

## **Data Models**

Hierarchical, Network, Relational, Object-Oriented.

## **DBMS Architecture**

1-Tier, 2-Tier, and 3-Tier architecture.

## **Keys**

Primary Key, Foreign Key, Candidate Key, Alternate Key, Composite Key, Super Key.

## **ER Model**

Entity, Attribute, Relationship, Cardinality.

## **Normalization**

1NF, 2NF, 3NF, BCNF to reduce redundancy.

## **SQL Introduction**

Structured Query Language used to interact with databases.

## **DDL Commands**

CREATE, ALTER, DROP, TRUNCATE.

## **DML Commands**

INSERT, UPDATE, DELETE.

## **DQL Commands**

SELECT.

## **DCL Commands**

GRANT, REVOKE.

## **TCL Commands**

COMMIT, ROLLBACK, SAVEPOINT.

## **Constraints**

NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK, DEFAULT.

## **Joins**

INNER, LEFT, RIGHT, FULL JOIN.

## **Functions**

Aggregate functions: COUNT, SUM, AVG, MIN, MAX.

## **Views**

Virtual tables based on SQL queries.

## **Indexes**

Improve search performance.

## **Transactions**

Logical unit of work.

## **ACID Properties**

Atomicity, Consistency, Isolation, Durability.

## **Concurrency Control**

Locking and timestamp techniques.

## **Deadlock**

Occurs when transactions wait for each other indefinitely.

## **Backup and Recovery**

Protects data from failures.

## **Distributed DBMS**

Database distributed across multiple locations.

## **NoSQL Databases**

MongoDB, Cassandra, Redis.

## **DBMS vs RDBMS**

RDBMS stores data in related tables and supports SQL.

## **Popular Databases**

MySQL, PostgreSQL, Oracle, SQL Server, TiDB.